

Java Quick Documentation

What is Java?

Java is a high-level, object-oriented, platform-independent language. Code is compiled to bytecode executed by the JVM.

Architecture

Source (.java) -> javac -> Bytecode (.class) -> JVM -> Operating System

Basic Concepts

Variables, data types, operators, conditionals, loops, methods.

Object-Oriented Programming

Classes, objects, constructors, encapsulation, inheritance, polymorphism, abstraction, interfaces.

Collections

List, Set, Map, Queue from java.util.

Exception Handling

try, catch, finally, throw, throws.

File Handling

Use java.io and java.nio packages.

Java 8+ Features

Lambda expressions, Stream API, Optional, Date-Time API.

JVM vs JRE vs JDK

JVM runs bytecode; JRE provides runtime; JDK includes compiler and tools.

Popular Frameworks

Spring Boot, Hibernate, JUnit, Maven, Gradle.